MITOCHONDRIAL DNA CHARACTERIZATION OF INDIGENOUS STRAINS OF COMMON CARP *(Cyprinus carpio)*

by

Write your full name here.

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of
Master of Science in Aquaculture and Aquatic Resources Management

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April 2002

AUTHOR'S DECLARATION

I, (your name), declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it is my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis, including final revisions.

Date: Click or tap to enter a date.

Name (in printed letters): SUBIN PINKAYAN

Signature:

ACKNOWLEDGMENTS

Type your acknowledgments here. This section is typically reserved for personal and professional dedications. You may also acknowledge your donor or funder here or include copyright acknowledgements of journals whose articles you have significantly incorporated in this thesis. Maintain one blank 1.5 line spacing between paragraphs.

[one page, maximum]

ABSTRACT

Your abstract should present a summary of the main sections of your thesis, research study or dissertation. It should be between 200 to 350 words, or not more than one page. Include also the most significant achievements of your study in order to demonstrate the value of your work.

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LIST OF ABBREVIATIONS

A = aquifer area

Ar = artificial ground water aquifer

C = expected value of objective function

CHAPTER 1
INTRODUCTION

1.1 Background of the Study

Start your paragraph here; justified, no indentation but space between the last paragraph and the next one. Maintain a 1.5 line spacing throughout. This is NOT a Literature Review section.

Each paragraph should have at least 4 sentences or 8 to 10 lines. Follow the appropriate structure of writing a clear paragraph. Consult your adviser about the subsections. Maintain one space between the last line of this section and the next subsection.

1.2 Statement of the Problem

Write approximately three paragraphs only. Each paragraph should have at least 4 sentences. Follow the appropriate structure of writing a clear paragraph.

1.3 [Research Questions- Discuss with your adviser.]

Introduce your research questions in one sentence followed by a bullet list, left-aligned.

1. Follow a 0.5-inch Tab setting
2.
3.

1.4 Objectives of the Study

Introduce your main objective first in one sentence, followed by a bullet list of specific objectives, left aligned.

1. Follow a 0.5-inch Tab setting.
2. Add more here.

1.5 Write the next section here.

Start your paragraph here.

1.6 Organization of the Study

Start your paragraph here.

CHAPTER 2
LITERATURE REVIEW

For clarity, you are recommended to include, at the beginning and at the end of each chapter, an introductory and a summary paragraph, respectively. This will provide contextual information that will guide your future readers, including any external examiners.

Example 2.1: Introductory paragraph

The idea of applying method X for the task of A has been well explored. In this chapter, an outline of the methodologies M, N, P and conclusions drawn from literature will be presented. The chapter will be divided into J sections, each of which is a discussion regarding aspect D, E, and F of the problem.

Example 2.2: Summary paragraph

Presented in this chapter are the concepts and approaches related to the problem of A. There are I methods to tackle problem A. [Brief overview of methods]. Among these methods, method M has the advantage of ____. Despite its advantage, method M comes with a drawback ___. It has been suggested by [citation list] that modification Y could improve method X. This has been supported by evidence [citation list]. The following chapter will outline the research design that implemented this modification.

Write your introductory paragraph/s to give an overview of the chapter (except for Chapter 1). Follow the appropriate structure of writing paragraphs. Paragraphs should have at least four sentences (8 lines). Maintain one blank line between paragraphs.

2.1 Heading, Level 2

This section presents some guidelines on how to create and format tables and figures following the APA Style with some examples. Every table and figure should serve a purpose. A table or figure can be referred to in the text by its number [e.g., As shown in Table 2.1…, as can be seen in the results of the testing (see Figure 2.1)]. Avoid writing "the table above" or "the figure below" as the position of a figure or table might change during the writing process.

Tables and figures can be generated in different ways using many programs. Table 2.1 presents the format of a table following the APA style. Use Quick Parts (in the Word search box, type "Quick Parts", and select "Explore Quick Parts") to insert Table and Figure Titles. Place table titles above the tables and figure captions below the figures. See FAQ for instruction on how to use Quick Parts. Separate the paragraph and the table or figure title by a double-spaced blank line. Titles should be brief, clear, and explanatory.

Repeat the column headings on the second page of the table (see Table 2.1). Separate this paragraph from the table by a double-spaced line. Tables and figures can be placed at the start or end of a page. Fit the table or figure between the margins and in one page.

As there is very little space left for the table on this page, present the table on the next page. You can add more content in this section. The description should be as close to the table or figure as possible.

Table 2.1. Means and Standard Deviations of Scores on Baseline Measures

Scale	High BAS group	Moderate BAS group	*p*
BAS-T	46.17 (2.87)	37.99 (1.32)	<.001
SR	17.94 (1.88)	11.52 (1.84)	<.001
BDI	7.11 (6.50)	6.18 (6.09)	.254
ASRM	6.46 (4.01)	5.63 (3.69)	.109
M-SRM	11.05 (3.36)	11.76 (2.75)	.078

2.2 Heading, Level 2

Add a short introductory sentence/s here.

2.2.1 Heading, Level 3

Start your paragraph here. The APA Publication Manual also provides examples of qualitative tables that present a summary of your data or show variable descriptions. **Consult your advisor first.** The formatting guidelines for these tables are similar to 2.1, Page 3.

Separate the paragraph and the table by a double-spaced blank line. Titles should be brief, clear, and explanatory.[1]

2.2.2 Heading, Level 3

(One space between the last line of this section and the next Level 2 heading)

2.3 Heading, Level 2

As for figures, the figure title should also be written in bold below the figure number (in bold) separated by a double-spaced blank line as shown in Figure 2.1.[2] The size and density of the elements in a figure must be considered when deciding on the font size and spacing.[3] Continue with the paragraph here.

Continue with another paragraph here. The table or figure should be as close to the description as possible or when it is first mentioned. Fit the tables and figures between the margins (1 inch on the left and 1 inch on the right margin). There should be one blank double-spaced line between the previous paragraph and the figure number, and between the figure number and the title as shown in the figure on Page 6.

---

[1] The table body may be single-spaced, one-and-a-half-spaced or double-spaced. Notes can be written in a smaller font size, if necessary, of not less than 10. You can also reduce the font size for the table body to fit everything on one page or one table. However, uniformity should be observed for all figures and tables.

[2] The same font should be used throughout the thesis, but the font size for figures can be between 10-14 making sure that the presentation is legible. Adjust the footnotes, if necessary, between 10-12 font size.

[3] This means that crowding in the image should be avoided in order to give enough space between words.

Figure 2.1. Strategic Engineering Framework

Note. The first step in Systems Engineering is to carefully consider the Architecture that the system should or could have. This results in an initial choice of concept that needs to be better understood in terms of its anticipated performance, cost, and risk profile. This is where Integrated Modeling and Simulation can be very helpful. Finally, we want to "fine tune" the design of the system so that it optimally meets the various objectives and constraints that have been laid out. Ideally, Multidisciplinary Design Optimization will yield an "optimal" design x^* at the end of this process.

(There should be one blank double-spaced line here.)

2.3.1 Heading, Level 3

This section presents an example of a block quotation of 40 words or more with in-text citations. It is indented 0.5 in from the left margin. A block quotation is written on a new line, and sometimes, with a narrative citation as shown in the following example. Do not use quotation marks. Quotation marks are only used when quoting a source within a text. Flores et al. (2018) described how they addressed potential research bias when working with people of color:

> Everyone on the research team belonged to a stigmatized group but also held privileged identities. Throughout the research process, we attended to the ways in which our privileged and oppressed identities may have influenced the research process, finding, and presentation of results. (p. 311)

Block quotations sometimes consist of two or more paragraphs. Indent the first line of each subsequent paragraph, an additional 0.5 in. The next example shows a block quotation with parenthetical citation consisting of two paragraphs.

> In order to accurately estimate whether people are likely to form positive or negative expectations on any given occasion, it is necessary to go beyond simply considering

chronic individual differences and identify the factors that make people more likely to form expectations in line with one bias or the other…

The present research sheds light on this issue by identifying a crucial distinction in the operation of these two trait biases in expectation formation. Specifically, people's valence weighting biases and self-beliefs about the future appear to shape expectations via qualitatively distinct processes. (Niese et al., 2019, p. 210)

The 7th edition of the APA Publication Manual states that interviews can refer to a published interview, personal interview/personal communication, or research participant interview. Formatting quotations from interviews follow the same guidelines as other quotations.

1. Present a quotation of fewer than 40 words in quotation marks within a text.
2. Present a quotation of 40 words or more in a block quotation indented below the text (2.3.1, Page 6).

Published interviews are accompanied by in-text citations following the APA style:

(Goldwater, 2020) or Goldwater (2020)

and a corresponding reference entry in the References section based on the standard format for the source (e.g., books, magazines, YouTube, etc.)

Goldwater, M. (2020, February 23). Dr Micah Goldwater discusses pseudoscience [Interview]. All in the mind with Sana Qadar: Suckers for pseudoscience. ABC Radio National. https://www.abc.net.au/radionational/Programs/allinthemind/suckers-for-pseudoscience/11976394

Quotations from research participant interviews do not have citations and are formatted differently from personal interviews/communications:

3. Mention in the text that the quotations are from the participant/s, as in this APA example (FG, block format):

   In focus group discussions, participants described their postretirement experiences, including the emotions associated with leaving work and its effective and practical implications. "Rafael" (64 years old, retired pilot) mentioned several difficulties associated with retirement, including feeling like he was "in a void without purpose . . . it took several months to develop new interests that motivated [him] each day."

Several other participants agreed, describing the entrance into retirement as "confusing," "lonely," "purposeless," and "boring." In contrast, others described the sense of "balance" and "relaxation" retirement brought to their lives.

4. You must conform to ethical agreements made with your participants with regard to confidentiality and/or anonymity when quoting them. Their consent must be obtained prior to including their information in the report.

Quotations from personal interviews/communications conducted informally to obtain additional information such as by email have no reference entry as readers cannot recover/find them. However, in-text citation must be added within the text or in the block format.

In-text citation: (J. Vargas, personal communication, November 20, 2012)

2.3.2 Heading, Level 3

Start your paragraph here.

***2.3.2.1* Heading, Level 4.** Capitalize the first letter of the main words and words of 4 letters or more, followed by a period. Start on the same line here and continue as a regular paragraph.

2.3.2.2 **Heading, Level 4.** Start on the same line here and continue as a regular paragraph.

2.3.2.2.1 ***Heading, Level 5.*** Capitalize the first letter of each main word and words of 4 letters or more, followed by a period. Start on the same line here and continue as a regular paragraph.

(1 space between the last line of this section and the next Level 2 heading)

2.4 Heading, Level 2

Add a short introductory sentence/s here.

2.4.1 Heading, Level 3

Start your paragraph here.

2.4.2 Heading, Level 3

Start your paragraph here.

***2.4.2.1* Heading, Level 4.** Capitalize the first letter of the main words and words of 4 letters or more, followed by a period. Start on the same line here and continue as a regular paragraph.

***2.4.2.2* Heading, Level 4.** Start on the same line here and continue as a regular paragraph.

2.4.2.2.1 ***Heading, Level 5.*** Capitalize the first letter of each main word and words of 4 letters or more, followed by a period. Start on the same line here and continue as a regular paragraph.

(1 space between the last line of this section and the next section heading)

2.5 Add more sections as required.

Start your paragraph here. Follow the format for headings and subheadings.

(1 space between the last line of this section and the next Level 2 heading)

2.6 Chapter Summary

CHAPTER 3
METHODOLOGY

For clarity, you are recommended to include, at the beginning and at the end of each chapter, an introductory and a summary paragraph, respectively. This will provide contextual information that will guide your future readers, including any external examiners.

CHAPTER 4
RESULTS AND DISCUSSION

Begin this chapter with an introductory paragraph that provides an overview of its content and purpose. Follow proper paragraph structure, each paragraph should contain at least four sentences (approximately eight lines). Maintain one blank line between paragraphs for clarity and readability.

Present all relevant results in a clear and logical order. Use *tables, graphs, and figures* to support your findings, ensuring that each is properly numbered, titled, and captioned. Describe the results objectively before providing any interpretation. Organize the presentation according to the study's research objectives, questions, or hypotheses to maintain coherence and focus.

After presenting all results, provide an integrated discussion in a separate section that explains their meaning and significance. Interpret the findings in relation to the study's objectives, the methods used, and the literature discussed in earlier chapters. Highlight any patterns, trends, or relationships observed in the data, and compare them with previous studies to identify similarities, differences, or new insights. Discuss possible reasons for observed results and emphasize how these contribute to understanding the research problem or advancing knowledge and practical applications in the field.

Avoid simply repeating data or restating information from earlier chapters. Instead, focus on critical analysis and synthesis of the findings. When applicable, acknowledge any limitations or uncertainties that may influence the interpretation of results.

Conclude the chapter with a brief summary highlighting the key findings and insights. This summary should provide a smooth transition to Chapter 5 (Conclusion and Recommendations), where the study's overall implications, conclusions, and suggested actions will be presented.

(Note: The specific structure of Chapter 4 may vary depending on the discipline or the preferences of your adviser and thesis committee. Use this guideline as a standard format and adjust it as necessary based on their feedback.)

CHAPTER 5
CONCLUSION AND RECOMMENDATIONS

This chapter may include sections such as: conclusions, recommendations, future research ideas, etc. that might be relevant to various sectors, be it research, industry, policy, or management. Example:

5.1 Conclusions

5.2 Recommendations

5.3 Future Research Directions

REFERENCES

Ahmann, E., Tuttle, L.J., Saviet, M., & Wright, S.D. (2018). A descriptive review of ADHD coaching research: Implication for college students. *Journal of Postsecondary Education and Disability, 31*(1), 17-39. https://www.ahead.org/professional-resources/publications/jped/archived-jped/jped-volume-31

Alonso-Tapia, J., Nieto, C., Merino-Tejedor, E., Huertas, J. A., & Ruiz, M. (2018). *Situated Goals Questionnaire for University Students (SGQ-U, CMS-U)* [Database record]. PsycTESTS. https://doi.org/10.1037/t66267-000

A separate file on how to format reference entries of different materials or sources is available on the LC website using the 7th edition of the APA Manual (2020).

(Placeholder1)

APPENDICES

APPENDIX A
TITLE

Different materials are presented in the APPENDICES. Label the materials in the order that they are mentioned in the text or section (e.g., "see Appendix A for the questions"). Large or oversized tables or figures that support, but are not important in the text, are included in the appendices in a portrait or landscape orientation. This section is for a single table, figure, image, or illustration.

All statistical summaries, tabulated outputs, and supporting visuals derived from the research data must be included in the Appendix. This includes:

1. Descriptive and inferential statistical results (e.g., summary tables, significance test outputs, and graphical representations).
2. Raw or summarized datasets that underpin the reported analyses, formatted clearly for reproducibility.
3. Photographs and documentation captured during fieldwork, experiments, or survey implementation.

These materials are required to enhance the transparency, traceability, and reproducibility of the study and to ensure that all data summaries and research process records be archived within the Appendix.

APPENDIX B
TITLE

This section is for multiple tables and / or figures. You can also write a short description of this section.

Table A1 Title

Add the table here with proper formatting style.

Table A2 Title

Add the table here with proper formatting style.

Figure A1 Title

Figure A2

VITA

This section presents a short description of the educational and professional achievements of the student.